THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A body continues in its state of rest or of uniform motion in a straight line unless acted on by an external force. This is: 1

(a) Newton's first law **(b)** the law of gravitation
(c) Newton's second law **(d)** Newton's third law

2. According to Newton's third law, action and reaction forces: 1

(a) are equal in magnitude, opposite in direction and act on different bodies
(b) cancel each other on the same body
(c) act on the same body
(d) are unequal

3. The weight of a body on the Moon compared to that on Earth is about: 1

(a) one-sixth **(b)** zero **(c)** the same **(d)** six times

4. Inertia of a body depends on its: 1

(a) colour **(b)** mass **(c)** speed only **(d)** volume

5. Action and reaction do not cancel because they: 1

(a) act on different bodies **(b)** act at different times
(c) are unequal **(d)** are in the same direction

6. Airbags in cars protect passengers by: 1

(a) stopping the car
(b) reducing the mass
(c) increasing the stopping time and reducing the force
(d) reducing the speed

7. The SI unit of force is: 1

(a) watt **(b)** pascal **(c)** newton **(d)** joule

8. A rocket moves upward because: 1

(a) the Earth attracts it
(b) the exhaust gases pushed downward exert an equal and opposite force on it
(c) air pushes it
(d) friction

9. A fielder draws his hands back while catching a fast ball to: 1

(a) increase the time of stopping and reduce the force
(b) stop the ball faster
(c) reduce the mass
(d) increase the force

10. 1 newton is the force that gives an acceleration of 1 m/s squared to a mass of: 1

(a) 10 kg **(b)** 1 g **(c)** 1 kg **(d)** 100 g

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State Newton's first law of motion. What is inertia? 2

12. Why does a passenger fall forward when a moving bus stops suddenly? 2

13. Why does a fielder draw his hands back while catching a fast ball? 2

14. State Newton's second law and write the formula. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A body of mass 2 kg is acted on by forces of 8 N east and 3 N west. Find the net force and acceleration. 3

16. A force of 12 N acts for 3 s on a body at rest of mass 4 kg. Find the velocity acquired. 3

17. Why do we need static friction to walk? What happens on ice? 3

OR

A car of mass 1000 kg moving at 20 m/s is stopped in 5 s. Find the force applied by the brakes. 3

18. Explain why the horse pulling a cart can make the cart move even though the cart pulls the horse back with an equal force. 3

OR

Derive F = ma from the second law in the form F = change in momentum/time. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. State Newton's three laws of motion and give one example of each. 5

OR

A body of mass 5 kg is moving on a rough surface with an applied force of 30 N and friction of 10 N. Find the acceleration, the velocity after 4 s and the distance covered in 4 s starting from rest. 5

20. Describe an activity to show the inertia of rest and of motion with a coin, a card and a glass or a pile of coins. 5

OR

Explain the concepts of mass, weight and the variation of g, with calculations of weight on Earth and Moon for a 50 kg astronaut. 5

END OF QUESTION PAPER — Total: 40 Marks